Reg. No. | | | | | | | | | | [13CE52]

FIFTH SEMESTER B.E/B.TECH EXAMINATION, DECEMBER - 2015

HIGHWAY ENGINEERING

Time: 3 Hours **Maximum Marks:70**

Instructions:
1. Answer any **FIVE** full Questions.
2. Draw neat diagram wherever necessary.

1. a. What are the important policies mentioned in the vision: 2021 document of Road Development plan. **08**

b. Explain: **06**

i. Indian Road Congress.

ii. Highway Research Board.

2. a. What are the engineering surveys to be carried out in fixing the highway alignment? Explain briefly. **08**

b. Define camber. What are the objectives of providing camber? Indicate the IRC recommended values of camber. **06**

3. a. What is super elevation? Enumerate the steps for practical design of superelevation. **06**

b. A national highway is passing through a rolling terrain in heavy rainfall area. It has a horizontal curve of radius of 400m. Design the length of transition curve assuming suitable data wherever necessary. **08**

4. a. List the factors in which SSD depends. Explain PIEV theory. **07**

b. Calculate the sight distance required for a design speed of 100kmph if the road is at grade 2.5% ascending and descending gradients and braking efficiency of vehicle is 50%. Also calculate the intermediate sight distance. **07**

5. a. The properties of subgrade soil are given below. **06**

Passing 0.074mm sieve = 70%

Liquid limit = 58%

Plasticity Index = 25%

Classify the soil by HRB system with G.I value also discuss the suitability of soil as subgrade material.

b. Explain briefly the desirable properties of road aggregates. List the various tests to be conducted on road aggregates. **08**

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6. a. Differentiate between flexible pavement and rigid pavement. **05**

b. Briefly explain: **09**

i. Penetration Macadam.

ii. Bituminous Macadam.

iii. Bituminous concrete.

7. a. Explain the significance of highway drainage system. **06**

b. What is highway maintenance? Explain various types of failures in rigid and flexible pavement. **08**

8. a. Write a short note on structural evaluation of pavement. **08**

b. Briefly explain the principle involved in FWD (Falling Weight Deflectometer) method **06**